Lloyd D'Souza

AI & ML ENGINEERING STUDENT

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SUMMARY

AI & ML engineering student who builds complete applications: the backend service, the checks around a model, the interface that explains its output, and the tests. Four projects built end to end, two team hackathons, and one hackathon conducted for my college.

EDUCATION

Bachelor of Engineering (BE) in Artificial Intelligence and Machine Learning
Mangalore Institute of Technology and Engineering (MITE) · CGPA 8.43 · expected graduation 2028

Pre-University Course (Science) — Christ King PU College · 86%

SSLC (Class 10) — Christ King English Medium School · 85%

PROJECTS

Trustworthy News Detector (TND) · v1.0.0 · github.com/lloydaq/TND-The-News-Detector
Python, FastAPI, Pydantic, Ollama, Trafilatura, HTTPX; HTML, CSS, JavaScript

- Extracts claims from an article with a local LLM, searches GNews and GDELT, and checks each claim against the evidence retrieved.
- Deterministic 0–100 risk score; each run is marked COMPLETE or DEGRADED with every failure listed; pytest and Playwright suites.

DataSentry · v0.1.0 · github.com/lloydaq/DataSentry
Python, FastAPI, pandas, NumPy, scikit-learn; HTML, CSS, JavaScript

- Profiles a CSV and screens it for target leakage, label conflicts and quality problems, reporting each finding with its evidence and limits.
- Cross-validated label review and train/test overlap checks; JSON and HTML reports; runs locally with no database or external calls.

FlowCast · v1.0.0 · live demo at lloydaq.github.io/flowcast
Python, pandas, LightGBM, PyTorch (DCRNN), FastAPI; React, TypeScript, Leaflet

- Forecasts traffic speed 15–60 min ahead for 207 LA highway sensors (METR-LA): four baselines, LightGBM and a DCRNN, on identical rows.
- Leakage-tested features, a day-block bootstrap to compare models, and a React dashboard that replays June 2012.

CampusSignal · Not deployed; demo data · github.com/lloydaq/CampusSignal-pro
Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL; React, TypeScript, Tailwind CSS

- A multi-college campus platform: notices, department pages, a moderated community, academic records and issue reporting.
- College isolation enforced by the server on every request; server-side sessions, CSRF checks, Argon2id, tenant-isolation tests.

HACKATHONS

JalSetu-Lite · Build for Billions Hackathon, NITK
Backend: verification, security and disputes

- Built backend workflows for order verification, delivery validation and dispute handling.
- Worked on OTP verification and server-side validation logic.
- Developed APIs for the water-delivery workflow with teammates.

FocusCut · Tech Fusion 2.0
AI-powered learning assistant

- Contributed to a tool that turns YouTube videos into summaries, key takeaways and flashcards.
- Worked on interactive timelines and progress tracking for long-form learning.

LEADERSHIP

AIvibeX · Mangalore Institute of Technology and Engineering
Conducted a 5-hour AI hackathon for the college

CERTIFICATION

Foundations of Virtual Reality · NPTEL, IIT Madras
Jul–Oct 2026, 12-week course · Elite certificate · consolidated score 67% · 4 credits recommended

SKILLS

Languages	Python, JavaScript, TypeScript, Java, C, HTML, CSS	**ML and data**	NumPy, pandas, scikit-learn, LightGBM, PyTorch, Ollama
Backend	FastAPI, REST APIs, Pydantic, PostgreSQL, SQLAlchemy	**Frontend**	React, Vite, Tailwind CSS
Tools	Git, GitHub, VS Code, pytest, Playwright, GitHub Actions	**Spoken**	English, Hindi, Kannada